UNIT II DESCRIBING DATA

Types of Data - Types of Variables -Describing Data with Tables and Graphs –Describing Data with Averages - Describing Variability - Normal Distributions and Standard (z) Scores

1. THREE TYPES OF DATA

- Any statistical analysis is performed on **data,** a *collection of actual observations or scores in a survey or an experiment.*
- The precise form of a statistical analysis often depends on whether data are qualitative, ranked, or quantitative.
- Generally, **qualitative data** consist of words (Yes or No), letters (Y or N), or numeri- cal codes (0 or 1) that represent a class or category.
- **Ranked data** consist of numbers (1st, 2nd, . . . 40th place) that represent relative standing within a group.
- **Quantitative data** consist of numbers (weights of 238, 170, . . . 185 lbs) that represent an amount ora count.
- To determine the type of data, focus on a single observation in any collection of observations.
- For example, the weights reported by 53 male students in Table 1.1 are quantitative data, since any single observation, such as 160 lbs, represents an amount of weight.

- If the weights in Table 1.1 had been replaced with ranks, beginning with a rank of 1 for the lightest weight of 133 lbs and ending with a rank of 53 for the heavi est weight of 245 lbs, these numbers would have been ranked data, since any single observation represents not an amount, but only relative standing within the group of 53students.
- Finally, the Y and N replies of students in Table 1.2 are qualitative data, since any single observation is a letter that represents a class of replies.

Table 1.1
QUANTITATIVE DATA: WEIGHTS (IN POUNDS) OF MALESTATISTICS STUDENTS

160	168	133	170	150	165	158	165
193	169	245	160	152	190	179	157
226	160	170	180	150	156	190	156
157	163	152	158	225	135	165	135
180	172	160	170	145	185	152	
205	151	220	166	152	159	156	
165	157	190	206	172	175	154	

Table 1.2
QUALITATIVE DATA: "DO YOU HAVE A FACEBOOKPROFILE?" YES (Y) OR NO (N) REPLIES OF STATISTICS STUDENTS

Y	Y	Y	N	N	Y	Y	Y
Y	Y	Y	N	N	Y	Y	Y
N	Y	N	Y	Y	Y	Y	Y
Y	Y	N	Y	N	Y	N	Y
Y	N	Y	N	N	Y	Y	Y
Y	Y	N	Y	Y	Y	Y	Y
N	N	N	N	Y	N	N	Y
Y	Y	Y	Y	Y	N	Y	N
Y	Y	Y	Y	N	N	Y	Y
N	Y	N	N	Y	Y	Y	Y
	Y	Y	N				

2. TYPES OF VARIABLES

A **variable** *is a characteristic or property that can take on different values.*

Constant
A characteristic or property that can take on only one value.

Discrete Variable
A variable that consists of isolated numbers separated by gaps.

Continuous Variable
A variable that consists of numbers whose values, at least in theory, have no restrictions.

Approximate Numbers
Numbers that are rounded off, as is always the case with values for continuous variables.

Approximate Numbers
Whenever values are rounded off, as is always the case with actual values for continuous variables, the resulting numbers are **approximate,** *never exact.*

Experiment
A study in which the investigator decides who receives the special treatment.

Independent Variable
The treatment manipulated by the investigator in an experiment.

Dependent Variable
A variable that is believed to have been influenced by the independent variable.

3. DESCRIBING DATA WITH TABLES

FREQUENCY DISTRIBUTIONS FOR QUANTITATIVE DATA

A *frequency distribution* is a collection of observations produced by sorting observations into classes and showing their frequency (f) of occurrence in each class.

GUIDELINES FOR FREQUENCY DISTRIBUTIONS

Essential

1. Each observation should be included in one, and only one, class.

Example: 130–139, 140–149, 150–159, etc. It would be incorrect to use 130–140, 140–150, 150–160, etc., in which, because the boundaries of classes overlap, an observation of 140 (or 150) could be assigned to either of two classes.

2. List all classes, even those with zero frequencies.

Example: Listed in Table 2.2 is the class 210–219 and its frequency of zero. It would be incorrect to skip this class because of its zero frequency.

3. All classes should have equal intervals.

Example: 130–139, 140–149, 150–159, etc. It would be incorrect to use 130–139, 140–159, etc., in which the second class interval (140–159) is twice as wide as the first class interval (130–139).

Optional

4. All classes should have both an upper boundary and a lower boundary.

Example: 240–249. Less preferred would be 240–above, in which no maximum value can be assigned to observations in this class.

5. Select the class interval from convenient numbers, such as 1, 2, 3, . . . 10, particularly 5 and 10 or multiples of 5 and 10.

Example: 130–139, 140–149, in which the class interval of 10 is a convenient number. Less preferred would be 130–142, 143–155, etc., in which the class interval of 13 is not a convenient number.

6. The lower boundary of each class interval should be a multiple of the class interval.

Example: 130–139, 140–149, in which the lower boundaries of 130, 140, are multiples of 10, the class interval. Less preferred would be 135–144, 145–154, etc., in which the lower boundaries of 135 and 145 are not multiples of 10, the class interval.

7. Aim for a total of approximately 10 classes.

Example: The distribution in Table 2.2 uses 12 classes.

Table 2.2
FREQUENCY DISTRIBUTION (GROUPED DATA)

WEIGHT	*f*
240–249	1
230–239	0
220–229	3
210–219	0
200–209	2
190–199	4
180–189	3
170–179	7
160–169	12
150–159	17
140–149	1
130–139	3
Total	53

Constructing Frequency Distributions

1. Find the range, that is, the difference between the largest and smallest observations. The range of weights in Table 1.1 is 245-133 = 112.
2. Find the class interval required to span the range by dividing the range by the desired number of classes. In the present example,

$$\text{Class interval} = \frac{\text{range}}{\text{desired number of classes}} = \frac{112}{10} = 11.2$$

3. Round off to the nearest convenient interval (such as 1, 2, 3, . . . 10, particularly 5 or 10 or multiples of 5 or 10). In the present example, the nearest convenient interval is 10.
4. Determine where the lowest class should begin. (Ordinarily, this number should be a multiple of the class interval.) In the present example, the smallest score is 133, and therefore the lowest class should begin at 130, since 130 is a multiple of 10 (the class interval).
5. Determine where the lowest class should end by adding the class interval to the lower boundary and then subtracting one unit of measurement. In the present example, add 10 to 130 and then subtract 1, the unit of measurement, to obtain 139—the number at which the lowest class should end.
6. Working upward, list as many equivalent classes as are required to include the largest observation. In the present example, list 130–139, 140–149, . . . , 240–249, so that the last class includes 245, the largest score.
7. Indicate with a tally the class in which each observation falls. For example, the first score in Table 1.1, 160, produces a tally next to 160–169; the next score, 193, produces a tally next to 190–199; and so on.
8. Replace the tally count for each class with a number—the frequency (f)—and show the total of all frequencies. (Tally marks are not usually shown in the final frequency distribution.)
9. Supply headings for both columns and a title for the table.

Outlier - A very extreme score.

Relative Frequency Distribution

A frequency distribution showing the frequency of each class as a fraction of the total frequency for the entire distribution.

Constructing Relative Frequency Distributions

To convert a frequency distribution into a relative frequency distribution, divide the frequency for each class by the total frequency for the entire distribution. Table 2.5 illustrates a relative frequency distribution based on the weight distribution of Table 2.2.

Table 2.5
RELATIVE FREQUENCY DISTRIBUTION

WEIGHT	*f*	RELATIVE *f*
240–249	1	.02
230–239	0	.00
220–229	3	.06
210–219	0	.00
200–209	2	.04
190–199	4	.08
180–189	3	.06
170–179	7	.13
160–169	12	.23
150–159	17	.32
140–149	1	.02
130–139	3	.06
Total	53	1.02*

Cumulative Frequency Distributions

***Cumulative frequency distributions* show the total number of observations in each class and in all lower-ranked classes.**

Constructing Cumulative Frequency Distributions

To convert a frequency distribution into a cumulative frequency distribution, add to the frequency of each class the sum of the frequencies of all classes ranked below it.

This gives the cumulative frequency for that class. Begin with the lowest-ranked class in the frequency distribution and work upward, finding the cumulative frequencies in ascending order. In **Table 2.6,** the cumulative frequency for the class 130–139 is 3, since there are no classes ranked lower.

Table 2.6
CUMULATIVE FREQUENCY DISTRIBUTION

WEIGHT	*f*	CUMULATIVE *f*	CUMULATIVE PERCENT
240–249	1	53	100
230–239	0	52	98
220–229	3	52	98
210–219	0	49	92
200–209	2	49	92
190–199	4	47	89
180–189	3	43	81
170–179	7	40	75
160–169	12	33	62
150–159	17	21	40
140–149	1	4	8
130–139	3	3	6
Total	53		

Cumulative Percentages

If relative standing within a distribution is particularly important, then cumulative frequencies are converted to cumulative percentages.

A glance at Table 2.6 reveals that 75 percent of all weights are the same as or lighter than the weights between 170 and 179 lbs.

To obtain this cumulative percentage (75%), the cumulative frequency of 40 for the class 170–179 should be divided by the total frequency of 53 for the entire distribution.

Percentile Rank of an Observation

Percentage of scores in the entire distribution with equal or smaller values than that score.

<u>Frequency Distributions For Qualitative (Nominal) Data</u>

- When, among a set of observations, any single observation is a word, letter, or numerical code, the data are qualitative.
- Simply determine the frequency with which observations occupy each class, and report these frequencies as shown in **Table 2.7** for the Facebook profile survey.
- This frequency distribution reveals that Yes replies are approximately twice as prevalent as No replies.

Ordered Qualitative Data

- It's totally arbitrary whether Yes is listed above or below No in Table 2.7.
- When, however, qualitative data have an ordinal level of measurement because observations can be ordered from least to most, that order should be preserved in the frequency table, as illustrated in **Table 2.8,** in which military ranks are listed in descending order from general to lieutenant.

Table 2.7
FACEBOOK PROFILE SURVEY

Response	*f*
Yes	56
No	27
Total	83

Table 2.8
RANKS OF OFFICERS IN THE U.S. ARMY (PROJECTED 2016)

RANK	*f*	PROPORTION	CUMULATIVE PERCENT
General	311	.004*	100.0
Colonel	13,156	.167	99.6
Major	16,108	.204	82.9
Captain	29,169	.370	62.5
Lieutenant	20,083	.255	25.5
Total	78,827		

Relative and Cumulative Distributions for Qualitative Data

- Frequency distributions for qualitative variables can always be converted into relative frequency distributions, as illustrated in Table 2.8.
- Furthermore, if measurement is ordinal because observations can be ordered from least to most, cumulative frequencies (and cumulative percentages) can be used.
- As illustrated in Table 2.8, it's appropriate to claim, for example, that a captain has an *approximate* percentile rank of 63 among officers since 62.5 (or 63) is the cumulative percent for this class.

4. DESCRIBING DATA WITH GRAPHS

Data can often be described even more vividly, particularly when you're attempting to communicate with a general audience, by converting frequency distributions into graphs.

Graphs For Quantitative Data

Histograms

A bar-type graph for quantitative data. The common boundaries between adjacent bars emphasize the continuity of the data, as with continuous variables.

Important features of histograms.

■ Equal units along the horizontal axis (the *X* axis, or abscissa) reflect the various class intervals of the frequency distribution.

■ Equal units along the vertical axis (the *Y* axis, or ordinate) reflect increases in frequency. (The units along the vertical axis do not have to be the same width as those along the horizontal axis.)

■ The intersection of the two axes defines the origin at which both numerical scales equal 0.

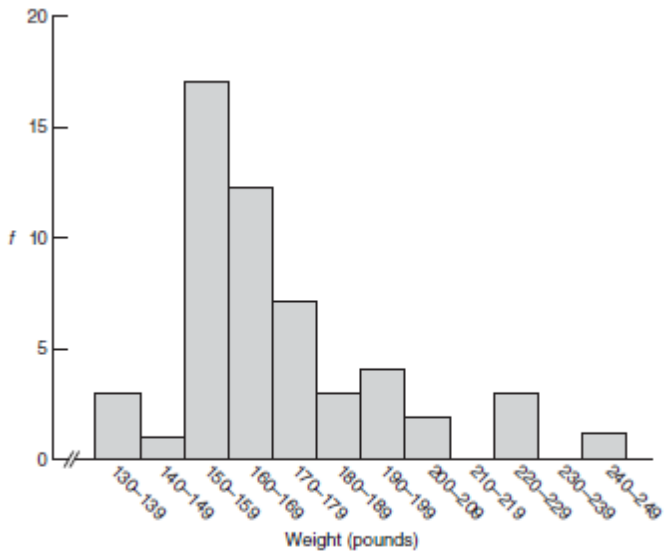

FIGURE 2.1
Histogram.

■ Numerical scales always increase from left to right along the horizontal axis and from bottom to top along the vertical axis. It is considered good practice to use wiggly lines to highlight breaks in scale, such as those along the horizontal axis in Figure 2.1, between the origin of 0 and the smallest class of 130–139.

■ The body of the histogram consists of a series of bars whose heights reflect the frequencies for the various classes. Notice that adjacent bars in histograms have common boundaries that emphasize the continuity of quantitative data for continuous variables. The introduction of gaps between adjacent bars would suggest an artificial disruption in the data more appropriate for discrete quantitative variables or for qualitative variables.

Frequency Polygon

A line graph for quantitative data that also emphasizes the continuity of continuous variables.

Step-by-step transformation of a histogram into a frequency polygon is described in panels A, B, C, and D of Figure 2.2.

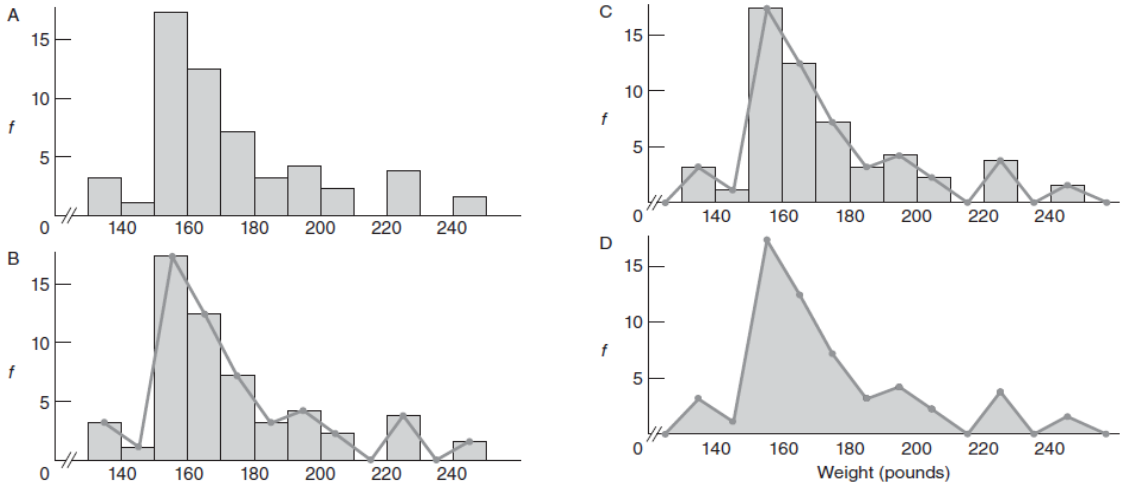

FIGURE 2.2 Transition from histogram to frequency polygon.

A. This panel shows the histogram for the weight distribution.
B. Place dots at the midpoints of each bar top or, in the absence of bar tops, at midpoints for classes on the horizontal axis, and connect them with straight lines. [To find the midpoint of any class, such as 160–169, simply add the two tabled boundaries (160 + 169 = 329) and divide this sum by 2 (329/2 = 164.5).]
C. Anchor the frequency polygon to the horizontal axis. First, extend the upper tail to the midpoint of the first unoccupied class (250–259) on the upper flank of the histogram. Then extend the lower tail to the midpoint of the first unoccupied class (120–129) on the lower flank of the histogram. Now all of the area under the frequency polygon is enclosed completely.
D. Finally, erase all of the histogram bars, leaving only the frequency polygon. Frequency polygons are particularly useful when two or more frequency distributions or relative frequency distributions are to be included in the same graph.

Stem and Leaf Display

A device for sorting quantitative data on the basis of leading and trailing digits.

Constructing a Display

The leftmost panel of **Table 2.9** re-creates the weights of the 53 male statistics students listed in Table 1.1.

To construct the stem and leaf display for these data, first note that, when counting by tens, the weights range from the 130s to the 240s.

Arrange a column of numbers, the stems, beginning with 13 (representing the 130s) and ending with 24 (representing the 240s).

Draw a vertical line to separate the stems, which represent multiples of 10, from the space to be occupied by the leaves, which represent multiples of 1.

Table 2.9
CONSTRUCTING STEM AND LEAF DISPLAY FROM WEIGHTS OF MALE STATISTICS STUDENTS

RAW SCORES					STEM AND LEAF DISPLAY
160	165	135	175		
193	168	245	165	13	3 5 5
226	169	170	185	14	5
152	160	156	154	15	2 7 1 7 8 0 2 0 2 6 9 8 2 6 4 7 6
180	170	160	179	16	0 3 5 8 9 0 0 0 6 5 5 5
205	150	225	165	17	2 0 0 0 2 5 9
163	152	190	206	18	0 0 5
157	160	159	165	19	3 0 0 0
151	190	172	157	20	5 6
157	150	190	156	21	
220	133	166	135	22	6 0 5
145	180	158		23	
158	152	152		24	5
172	170	156			

Selection of Stems

- Stem values are not limited to units of 10. Depending on the data, you might identify the stem with one or more leading digits that culminates in some variation on a stem value of 10, such as 1, 100, 1000, or even .1, .01, .001, and so on.
- For instance, an annual income of $23,784 could be displayed as a stem of 23 (thousands) and a leaf of 784. (Leaves consisting of two or more digits, such as 784, are separated by commas.)
- Stem and leaf displays represent statistical bargains. Just a few minutes of work produces a description of data that is both clear and complete.
- Even though rarely appearing in published reports, stem and leaf displays often serve as the first step toward organizing data.

Typical Shapes

Whether expressed as a histogram, a frequency polygon, or a stem and leaf display, an important characteristic of a frequency distribution is its shape. **Figure 2.3** shows some of the more typical shapes for smoothed frequency polygons.

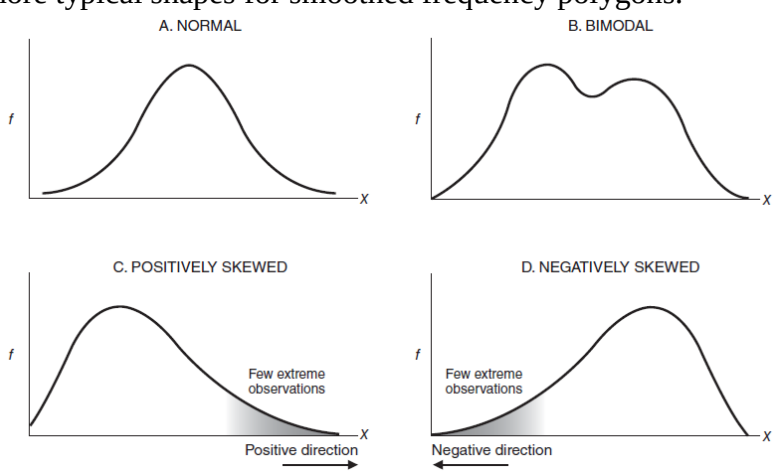

FIGURE 2.3 *Typical shapes.*

Positively Skewed Distribution

A distribution that includes a few extreme observations in the positive direction (to the right of the majority of observations).

Negatively Skewed Distribution

A distribution that includes a few extreme observations in the negative direction (to the left of the majority of observations).

CONSTRUCTING GRAPHS

1. Decide on the appropriate type of graph, recalling that histograms and frequency polygons are appropriate for quantitative data, while bar graphs are appropriate for qualitative data and also are sometimes used with discrete quantitative data.

2. Draw the horizontal axis, then the vertical axis, remembering that the vertical axis should be about as tall as the horizontal axis is wide.

3. Identify the string of class intervals that eventually will be superimposed on the horizontal axis. For qualitative data or ungrouped quantitative data, this is easy—just use the classes suggested by the data. For grouped quantitative data, proceed as if you were creating a set of class intervals for a frequency distribution.

4. Superimpose the string of class intervals (with gaps for bar graphs) along the entire length of the horizontal axis. For histograms and frequency polygons, be prepared for some trial and error—use a pencil! Do not use a string of empty class intervals to bridge a sizable gap between the origin of 0 and the smallest class interval. Instead, use wiggly lines to signal a break in scale, then begin with the smallest class interval. Also, do not clutter the horizontal scale with excessive numbers—use just a few convenient numbers.

5. Along the entire length of the vertical axis, superimpose a progression of convenient numbers, beginning at the bottom with 0 and ending at the top with a number as large as or slightly larger than the maximum observed frequency. If there is a considerable gap between the origin of 0 and the smallest observed frequency, use wiggly lines to signal a break in scale.

6. Using the scaled axes, ***construct bars (or dots and lines) to reflect the frequency of observations within each class interval.*** For frequency polygons, dots should be located above the midpoints of class intervals, and both tails of the graph should be anchored to the horizontal axis.

7. Supply labels for both axes and a title (or even an explanatory sentence) for the graph.

5. DESCRIBING DATA WITH AVERAGES

MODE

The value of the most frequent score.

Bimodal

Describes any distribution with two obvious peaks.

Example: Determine the mode for the following retirement ages: 60, 63, 45, 63, 65, 70, 55, 63, 60, 65, 63.

Answer: 63

MEDIAN

The median reflects the middle value when observations are ordered from least to most.

The median splits a set of ordered observations into two equal parts, the upper and lower halves. In other words, the median has a percentile rank of 50, since observations with equal or smaller values constitute 50 percent of the entire distribution.

Table 3.2
FINDING THE MEDIAN

A. INSTRUCTIONS

1. Order scores from least to most.
2. Find the middle position by adding one to the total number of scores and dividing by 2.
3. *If the middle position is a whole number*, as in the left-hand panel below, use this number to *count* into the set of ordered scores.
4. The value of the median equals the value of the score located at the middle position.
5. *If the middle position is not a whole number,* as in the right-hand panel below, use the two nearest whole numbers to *count* into the set of ordered scores.
6. The value of the median equals the value midway between those of the two middlemost scores; to find the midway value, add the two given values and divide by 2.

B. EXAMPLES

Set of five scores:	Set of six scores:
2, 8, 2, 7, 6	3, 8, 9, 3, 1, 8
1 2, 2, 6, 7, 8	1 1, 3, 3, 8, 8, 9
2 $\frac{5+1}{2}=3$	2 $\frac{6+1}{2}=3.5$
2, 2, 6, 7, 8	
3 1, 2, 3	
4 median = 6	1, 3, 3, 8, 8, 9
	5 1, 2, 3, 4
	6 median $=\frac{3+8}{2}=5.5$

MEAN

The mean is found by adding all scores and then dividing by the number of scores. That is,

$$Mean = \frac{sum\ of\ all\ scores}{number\ of\ scores}$$

Population - *A complete set of scores.*
Sample - *A subset of scores.*
Sample Mean (X)

The balance point for a sample, found by dividing the sum for the values of all scores in the sample by the number of scores in the sample.

Sample Size (n)

The total number of scores in the sample.

Formula for Sample Mean

SAMPLE MEAN

$$\bar{X} = \frac{\sum X}{n}$$

Population Mean (μ)

The balance point for a population, found by dividing the sum for all scores in the population by the number of scores in the population.

Population Size (N)

The total number of scores in the population.

Formula for Population Mean

POPULATION MEAN

$$\mu = \frac{\sum X}{N}$$

Distribution Is Not Skewed

When a distribution of scores is not too skewed, the values of the mode, median, and mean are similar, and any of them can be used to describe the central tendency of the distribution.

Differences between Mean and Median

Appreciable differences between the values of the mean and median signal the presence of a skewed distribution.

If the **mean exceeds the median,** the underlying **distribution is positively skewed** because of one or more scores with relatively large values.

If the **median exceeds the mean**, the underlying **distribution is negatively skewed** because of one or more scores with relatively small values.

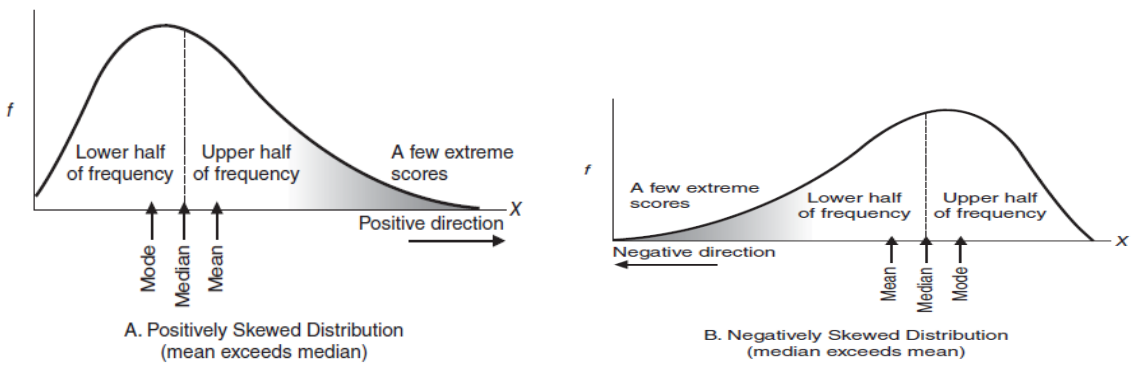

FIGURE 3.2 *Mode, median, and mean in positively and negatively skewed distributions.*

6. DESCRIBING VARIABILITY

Range *- The difference between the largest and smallest scores.*
Variance *- The mean of all squared deviation scores.*
Standard Deviation

The square root of the mean of all squared deviations from the mean, that is,

$$standard\ deviation = \sqrt{variance}$$

Sum of Squares (SS) *- The sum of squared deviation scores.*

Sum of Squares Formulas for Population

The definition formula provides the most accessible version of the population sum of squares:

SUM OF SQUARES (*SS*) FOR POPULATION (DEFINITION FORMULA)

$$SS = \sum(X-\mu)^2 \qquad (4.1)$$

SUM OF SQUARES (*SS*) FOR POPULATION (COMPUTATION FORMULA)

$$SS = \sum X^2 - \frac{(\sum X)^2}{N} \qquad (4.2)$$

VARIANCE:

$$variance = \frac{sum\ of\ all\ squared\ deviation\ scores}{number\ of\ scores}$$

or, in symbols:

VARIANCE FOR POPULATION

$$\sigma^2 = \frac{SS}{N}$$

STANDARD DEVIATION (σ) FOR POPULATION

A rough measure of the average amount by which scores in the population deviate on either side of their population mean.

STANDARD DEVIATION FOR POPULATION

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{SS}{N}}$$

Table 4.1
CALCULATION OF POPULATION STANDARD DEVIATION σ (DEFINITION FORMULA)

A. COMPUTATION SEQUENCE

Assign a value to N **1** representing the number of X scores
Sum all X scores **2**
Obtain the mean of these scores **3**
Subtract the mean from each X score to obtain a deviation score **4**
Square each deviation score **5**
Sum all squared deviation scores to obtain the sum of squares **6**
Substitute numbers into the formula to obtain population variance, σ^2 **7**
Take the square root of σ^2 to obtain the population standard deviation, σ **8**

Use sum of squares (definition formula)

B. DATA AND COMPUTATIONS

X	**4** $X-\mu$	**5** $(X-\mu)^2$
13	3	9
10	0	0
11	1	1
7	–3	9
9	–1	1
11	1	1
9	–1	1

1 $N = 7$ **2** $\Sigma X = 70$ **6** $SS = \Sigma (X-\mu)^2 = 22$

3 $\mu = \frac{70}{7} = 10$

7 $\sigma^2 = \frac{SS}{N} = \frac{22}{7} = 3.14$ **8** $\sigma = \sqrt{\frac{SS}{N}} = \sqrt{\frac{22}{7}} = \sqrt{3.14} = 1.77$

Table 4.2
CALCULATION OF POPULATION STANDARD DEVIATION (σ) (COMPUTATION FORMULA)

A. COMPUTATIONAL SEQUENCE

Assign a value to N representing the number of X scores **1**
Sum all X scores **2**
Square the sum of all X scores **3**
Square each X score **4**
Sum all squared X scores **5**
Substitute numbers into the formula to obtain the sum of squares, SS **6**
Substitute numbers into the formula to obtain the population variance, σ^2 **7**
Take the square root of σ^2 to obtain the population standard deviation, σ **8**

Use sum of squares (computation formula)

B. DATA AND COMPUTATIONS

X	**4** X^2
13	169
10	100
11	121
7	49
9	81
11	121
9	81

1 $N = 7$ **2** $\sum X = 70$ **5** $\sum X^2 = 722$

3 $(\sum X)^2 = 4900$

6 $SS = \sum X^2 - \frac{(\sum X)^2}{N} = 722 - \frac{4900}{7} = 722 - 700 = 22$

7 $\sigma^2 = \frac{SS}{N} = \frac{22}{7} = 3.14$ **8** $\sigma = \sqrt{\frac{SS}{N}} = \sqrt{\frac{22}{7}} = \sqrt{3.14} = 1.77$

Sum of Squares Formulas for Sample

Sample notation can be substituted for population notation in the above two formulas without causing any essential changes:

SUM OF SQUARES (*SS*) FOR SAMPLE (DEFINITION FORMULA)

$$SS = \Sigma(X - \overline{X})^2 \qquad (4.3)$$

(COMPUTATION FORMULA)

$$SS = \Sigma X^2 - \frac{(\Sigma X)^2}{n} \qquad (4.4)$$

STANDARD DEVIATION FOR SAMPLE (*s*)

A rough measure of the average amount by which scores in the sample deviate on either side of their sample mean.

VARIANCE FOR SAMPLE

$$s^2 = \frac{SS}{n-1}$$

STANDARD DEVIATION FOR SAMPLE

$$s = \sqrt{s^2} = \sqrt{\frac{SS}{n-1}}$$

where *s*2 and *s* represent the sample variance and **sample standard deviation,** *SS* is the sample sum of squares.

Table 4.3
CALCULATION OF SAMPLE STANDARD DEVIATION (S)
(DEFINITION FORMULA)

A. COMPUTATION SEQUENCE

Assign a value to n **1** representing the number of X scores
Sum all X scores **2**
Obtain the mean of these scores **3**
Subtract the mean from each X score to obtain a deviation score **4**
Square each deviation score **5**
Sum all squared deviation scores to obtain the sum of squares **6**
Substitute numbers into the formula to obtain the sample variance, s^2 **7**
Take the square root of s^2 to obtain the sample standard deviation, s **8**

Use sum of squares(definition formula)

B. DATA AND COMPUTATIONS

X	**4** $X - \bar{X}$	**5** $(X - \bar{X})^2$
7	4	16
3	0	0
1	−2	4
0	−3	9
4	1	1

1 $n = 5$ **2** $\Sigma X = 15$ **6** $SS = \Sigma(X - \bar{X})^2 = 30$

3 $\bar{X} = \frac{15}{5} = 3$

7 $s^2 = \frac{SS}{n-1} = \frac{30}{4} = 7.50$ **8** $s = \sqrt{\frac{SS}{n-1}} = \sqrt{\frac{30}{4}} = \sqrt{7.50} = 2.74$

Table 4.4
CALCULATION OF SAMPLE STANDARD DEVIATION (S)
(COMPUTATION FORMULA)

A. COMPUTATIONAL SEQUENCE

Assign a value to n representing the number of X scores **1**
Sum all X scores **2**
Square the sum of all X scores **3**
Square each X score **4**
Sum all squared X scores **5**
Substitute numbers into the formula to obtain the sum of squares, SS **6**
Substitute numbers into the formula to obtain the sample variance, s^2 **7**
Take the square root of s^2 to obtain the sample standard deviation, s **8**

Use sum of squares (computation formula)

B. DATA AND COMPUTATIONS

X	**4** X^2
7	49
3	9
1	1
0	0
4	16

1 $n = 5$ **2** $\Sigma X = 15$ **5** $\Sigma X^2 = 75$
3 $(\Sigma X)^2 = 225$

6 $SS = \Sigma X^2 - \frac{(\Sigma X)^2}{n} = 75 - \frac{225}{5} = 75 - 45 = 30$

7 $s^2 = \frac{SS}{n-1} = \frac{30}{4} = 7.50$ **8** $s = \sqrt{\frac{SS}{n-1}} = \sqrt{\frac{30}{4}} = \sqrt{7.50} = 2.74$

DEGREES OF FREEDOM (*df*)

Degrees of freedom (df) refers to the number of values that are free to vary, given one or more mathematical restrictions, in a sample being used to *estimate* a population characteristic.

INTERQUARTILE RANGE (IQR)

- The most important spinoff of the range, the **interquartile range (IQR),** *is simply the range for the middle 50 percent of the scores.*
- More specifically, the IQR equals the distance between the third quartile (or 75th percentile) and the first quartile (or 25th percentile), that is, after the highest quarter (or top 25 percent) and the lowest quarter (or bottom 25 percent) have been trimmed from the original set of scores.
- Since most distributions are spread more widely in their extremities than their middle, the IQR tends to be less than half the size of the range.

Table 4.6
CALCULATION OF THE IQR

A. INSTRUCTIONS

1. Order scores from least to most.
2. To determine how far to penetrate the set of ordered scores, begin at either end, then add 1 to the total number of scores and divide by 4. If necessary, round the result to the nearest whole number.
3. Beginning with the largest score, count the requisite number of steps (calculated in step 2) into the ordered scores to find the location of the third quartile.
4. The third quartile equals the value of the score at this location.
5. Beginning with the smallest score, again count the requisite number of steps into the ordered scores to find the location of the first quartile.
6. The first quartile equals the value of the score at this location.
7. The IQR equals the third quartile minus the first quartile.

B. EXAMPLE

1. 7, 9, 9, 10, 11, 11, 13
2. $(7 + 1)/4 = 2$
3. 7, 9, 9, 10, 11, 11, 13

   ↑

   2 1

4. third quartile = 11
5. 7, 9, 9, 10, 11, 11, 13

   ↑

   1 2

6. first quartile = 9
7. IQR = 11 − 9 = 2

7. Normal Distributions and Standard (*z*) Scores

Normal Curve

A theoretical curve noted for its symmetrical bell-shaped form.

Properties of the Normal Curve

■ Obtained from a mathematical equation, the **normal curve** is a theoretical curve defined for a continuous variable, and noted for its symmetrical bell-shaped form, as revealed in Figure 5.2.
■ Because the normal curve is symmetrical, its lower half is the mirror image of its upper half.
■ Being bell shaped, the normal curve peaks above a point midway along the horizontal spread and then tapers off gradually in either direction from the peak (without actually touching the horizontal axis, since, in theory, the tails of a normal curve extend infinitely far).
■ The values of the mean, median (or 50th percentile), and mode, located at a point midway along the horizontal spread, are the same for the normal curve.

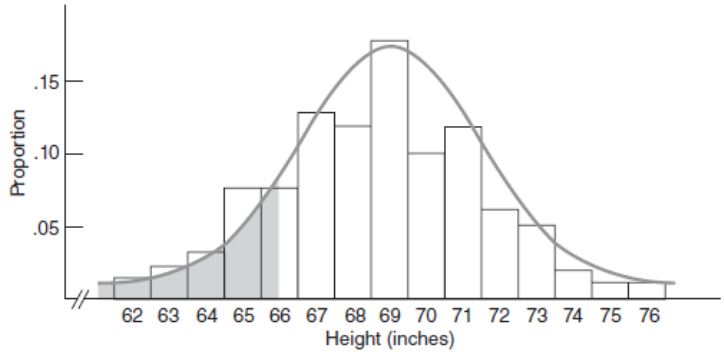

FIGURE 5.2 *Normal curve superimposed on the distribution of heights.*

***z* SCORES**

A *z* score is a unit-free, standardized score that, regardless of the original units of measurement, indicates how many standard deviations a score is above or below the mean of its distribution.

z SCORE

$$z = \frac{X - \mu}{\sigma}$$

A *z* score consists of two parts:

1. *a positive or negativ*e sign indicating whether it's above or below the mean; and

2. *a number* indicating the size of its deviation from the mean in standard deviation units.

- A *z* score of 2.00 always signifies that the original score is exactly two standard deviations above its mean.
- Similarly, a *z* score of –1.27 signifies that the original score is exactly 1.27 standard deviations below its mean. A *z* score of 0 signifies that the original score coincides with the mean.

Converting to *z* Scores

To answer the question about eligible FBI applicants, replace *X* with 66 (the maximum permissible height), μ with 69 (the mean height), and σ with 3 (the standard deviation of heights) and solve for *z* as follows:

<u>STANDARD NORMAL CURVE</u>

The tabled normal curve for z scores, with a mean of 0 and a standard deviation of 1.

$$\text{Mean of } z = \frac{X - \mu}{\sigma} = \frac{\mu - \mu}{\sigma} = \frac{0}{\sigma} = 0$$

$$\text{Standard deviation of } z = \frac{X - \mu}{\sigma} = \frac{\mu + 1\sigma - \mu}{\sigma} = \frac{1\sigma}{\sigma} = 1$$

Although there is an infinite number of different normal curves, each with its own mean and standard deviation, there is only one standard normal curve, with a mean of 0 and a standard deviation of 1.

Standard Normal Table

- Essentially, the standard normal table consists of columns of *z* scores coordinated with columns of proportions.
- In a typical problem, access to the table is gained through a *z* score, such as –1.00, and the answer is read as a proportion.

Using the Top Legend of the Table

- **Table 5.1** shows an abbreviated version of the standard normal curve.
- Notice that columns are arranged in sets of three, designated as A, B, and C in the legend at the top of the table.
- When using the top legend, all entries refer to the upper half of the standard normal curve.
- The entries in column A are *z* scores, beginning with 0.00 and ending (in the full-length table of Appendix C) with 4.00. Given a *z* score of zero or more, columns B and C indicate how the *z* score splits the area in the upper half of the normal curve.
- As suggested by the shading in the top legend, column B indicates the proportion of area between the mean and the *z* score, and column C indicates the proportion of area beyond the *z* score, in the upper tail of the standard normal curve.

Using the Bottom Legend of the Table

- Because of the symmetry of the normal curve, the entries in Table 5.1 and Table A of
- Now the columns are designated as A′ , B′ , and C′ in the legend at the bottom of the table.
- When using the bottom legend, all entries refer to the lower half of the standard normal curve.
- Imagine that the nonzero entries in column A′ are negative *z* scores, beginning with –0.01 and ending with –4.00. Given a negative *z* score, columns B′ and C′ indicate how that *z* score splits the lower half of the normal curve.
- As suggested by the shading in the bottom legend of the table, column B′ indicates the proportion of area between the mean and the negative *z* score, and column C′ indicates the proportion of area beyond the negative *z* score, in the lower tail of the standard normal curve.

TABLE 5.1
PROPORTIONS (OF AREAS) UNDER THE STANDARD NORMAL CURVE FOR VALUES OF z (FROM TABLE A OF APPENDIX C)

A z	B	C	A z	B	C	A z	B	C
0.00	.0000	.5000	0.40	.1554	.3446	0.80	.2881	.2119
0.01	.0040	.4960	0.41	.1591	.3409	0.81	.2910	.2090
•	•	•	•	•	•	•	•	•
						•	•	•
						•	•	•
						•	•	•
•	•	•	•	•	•	•	•	•
						0.99	.3389	.1611
						1.00	.3413	.1587
•	•	•	•	•	•	1.01	.3438	.1562
						•	•	•
						•	•	•
•	•	•	•	•	•	•	•	•
0.38	.1480	.3520	0.78	.2823	.2711	1.18	.3810	.1190
0.39	.1517	.3483	0.79	.2852	.2148	1.19	.3830	.1170
–z A′	B′	C′	–z A′	B′	C′	–z A′	B′	C′

FINDING PROPORTIONS

Example: Finding Proportions for *One* Score

Now we'll use a step-by-step procedure, adopted throughout this chapter, to find the proportion of all FBI applicants who are shorter than exactly 66 inches, given that the distribution of heights approximates a normal curve with a mean of 69 inches and a standard deviation of 3 inches.

(In the classic movie The President's Analyst, the director of the Federal Bureau of Investigation, rather short himself, encourages the recruitment of similarly short FBI agents. If, in fact, FBI agents are to be selected only from among applicants who are no taller than exactly 66 inches)

1. Sketch a normal curve and shade in the target area, as in the left part of **Figure 5.6**. Being less than the mean of 69, 66 is located to the left of the mean. Furthermore, since the unknown proportion represents those applicants who are shorter than 66 inches, the shaded target sector is located to the left of 66.

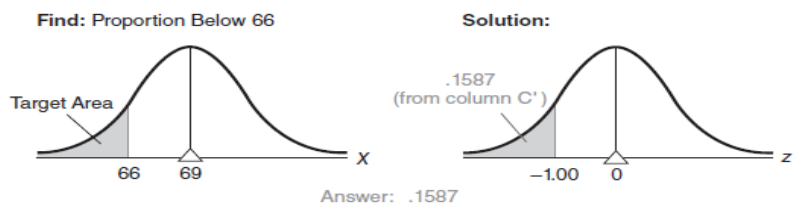

FIGURE 5.6
Finding proportions.

2. Plan your solution according to the normal table. Decide precisely how you will find the value of the target area. In the present case, the answer will be obtained from column C′ of the standard normal table, since the target area coincides with the type of area identified with column C′, that is, the area in the lower tail beyond a negative *z*.

3. Convert *X* to *z*. Express 66 as a *z* score:

$$z = \frac{X - \mu}{\sigma} = \frac{66 - 69}{3} = \frac{-3}{3} = -1$$

4. Find the target area. Refer to the standard normal table, using the bottom legend, as the *z* score is negative. The arrows in Table 5.1 show how to read the table. Look up column A' to 1.00 (representing a *z* score of –1.00), and note the corresponding proportion of .1587 in column C': This is the answer, as suggested in the right part of Figure 5.6. It can be concluded that only .1587 (or .16) of all of the FBI applicants will be shorter than 66 inches.

<u>Example: Finding Proportions *between* Two Scores</u>

Assume that, when not interrupted artificially, the gestation periods for human fetuses approximate a normal curve with a mean of 270 days (9 months) and a standard deviation of 15 days. What proportion of gestation periods will be between 245 and 255 days?

1. Sketch a normal curve and shade in the target area, as in the top panel of **Figure 5.7**. Satisfy yourself that, in fact, the shaded area represents just those gestation periods between 245 and 255 days.

2. Plan your solution according to the normal table. This type of problem requires more effort to solve because the value of the target area cannot be read directly from Table A. As suggested in the bottom two panels of Figure 5.7, the basic idea is to identify the target area with the difference between two overlapping areas whose values can be read from column C′ of Table A. The larger area (less than 255 days) contains two sectors: the target area (between 245 and 255 days) and a remainder (less than 245 days). The smaller area contains only the remainder (less than 245 days). Subtracting the smaller area (less than 245 days) from the larger area (less than 255 days), therefore, eliminates the common remainder (less than 245 days), leaving only the target area (between 245 and 255 days).

3. Convert *X* to *z* by expressing 255 as

$$z = \frac{255 - 270}{15} = \frac{-15}{15} = -1.00$$

and by expressing 245 as

$$z = \frac{245 - 270}{15} = \frac{-25}{15} = -1.67$$

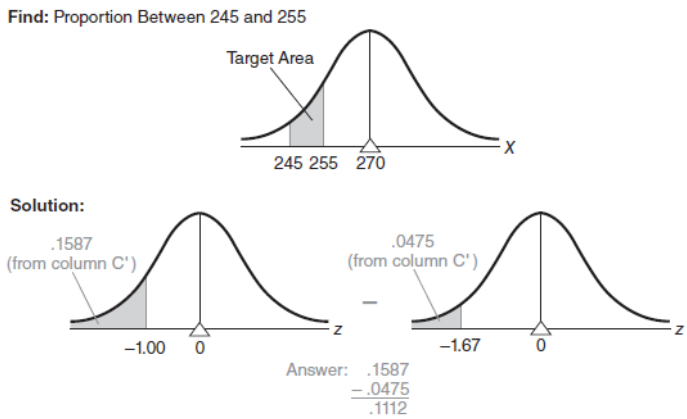

FIGURE 5.7
Finding proportions.

4. Find the target area. Look up column A′ to a negative *z* score of –1.00 (remember, you must imagine the negative sign), and note the corresponding proportion of .1587 in column C′. Likewise, look up column A′ to a *z* score of –1.67, and note the corresponding proportion of .0475 in column C′. Subtract the smaller proportion from the larger proportion to obtain the answer, .1112. Thus, only .11, or 11 percent, of all gestation periods will be between 245 and 255 days.